KEYLESS ENTRY SYSTEM

id091400109400

Outline

- A keyless entry system has been adopted in which locking/locking is performed by operation of the remote transmitter button, even from a location away front the vehicle.

Function

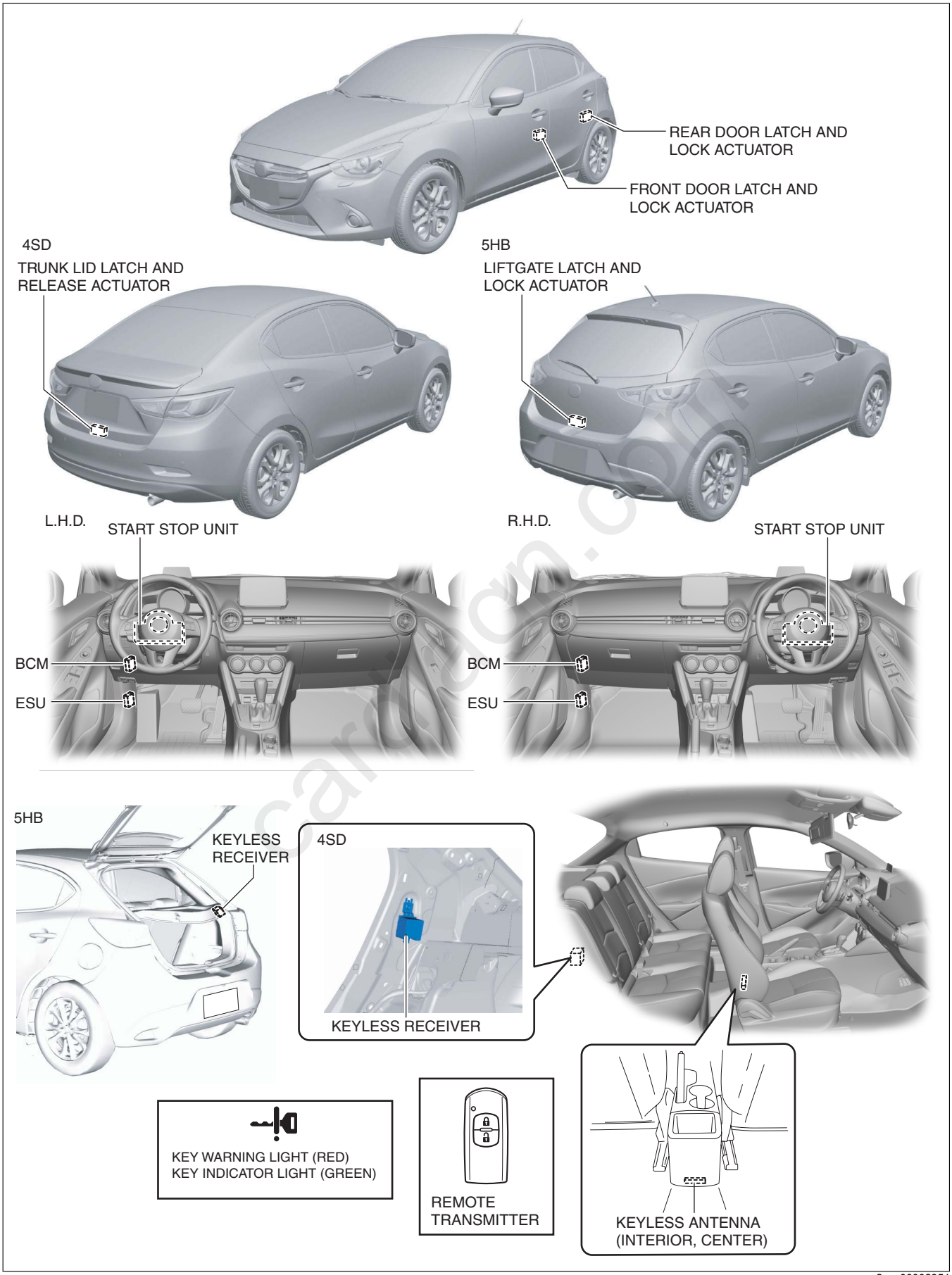
- When the following operation is performed, the start stop unit verifies the remote transmitter ID data.
  - — Locking/unlocking using a remote transmitter
  - — Trunk lid opening using remote transmitter (4SD)
- Each light flashes to notify the user that the lock/unlock operation is completed.
- The keyless entry system has the following functions in the event that the user operates the system incorrectly or to prevent the user from leaving the vehicle without locking.
  - — Auto re-lock function: After unlocking, if all doors or the liftgate/trunk lid are not opened and a period of time according to the personalized setting has elapsed, all doors are automatically locked.
  - — Remote transmitter pause function: All the functions of the remote transmitters inside the vehicle, other than the remote transmitter used for the door lock are paused.
  - — Warning function: Warns that there is a malfunction in the keyless entry system or the user operates the system incorrectly using any of the following:
    - Keyless warning alarm
    - KEY warning light (red)

Remote transmitter battery voltage low indication function

- This function notifies the driver that the battery voltage of the remote transmitter is low by flashing the KEY indicator light (green).

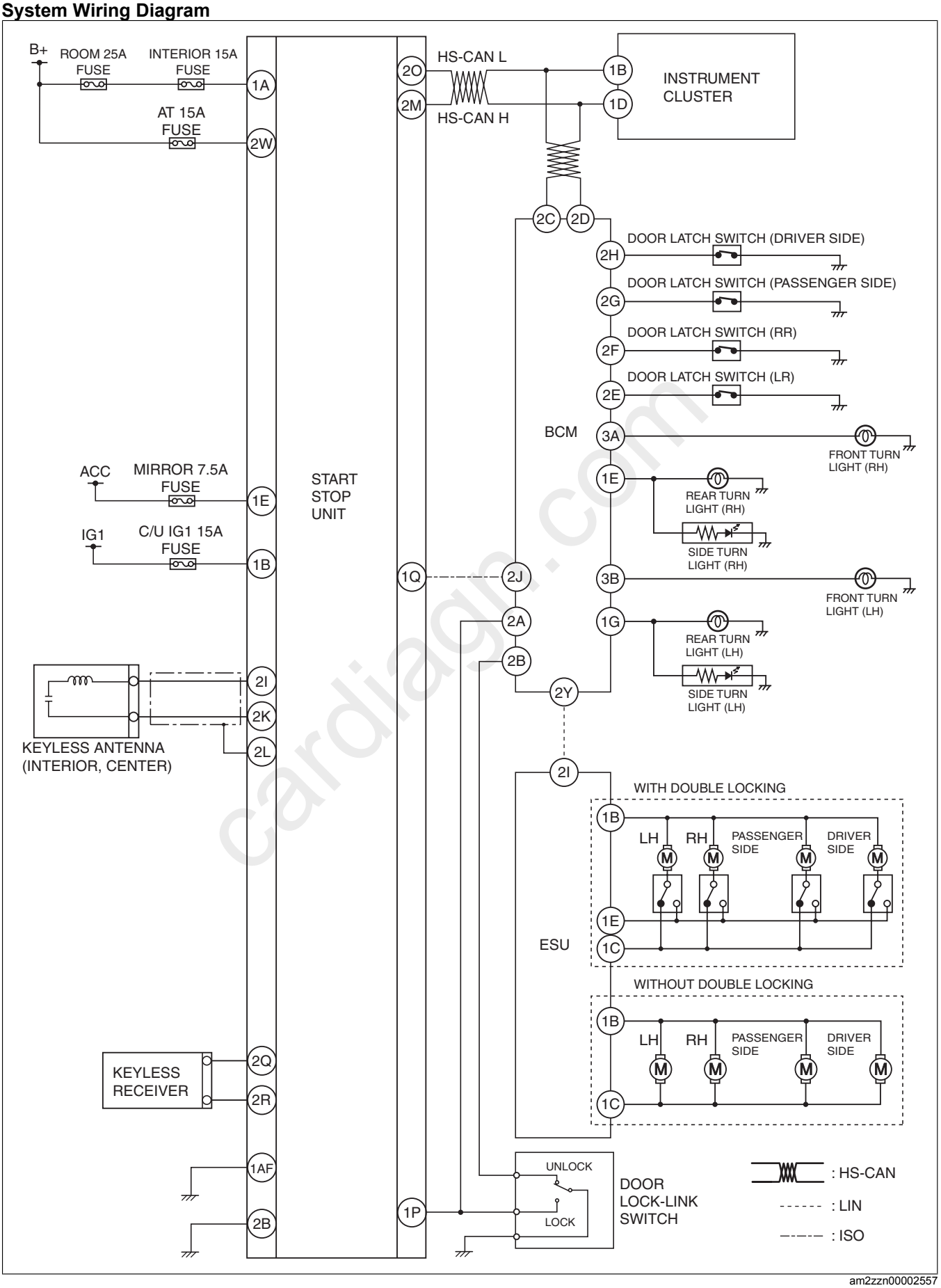
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Structural View



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System Wiring Diagram



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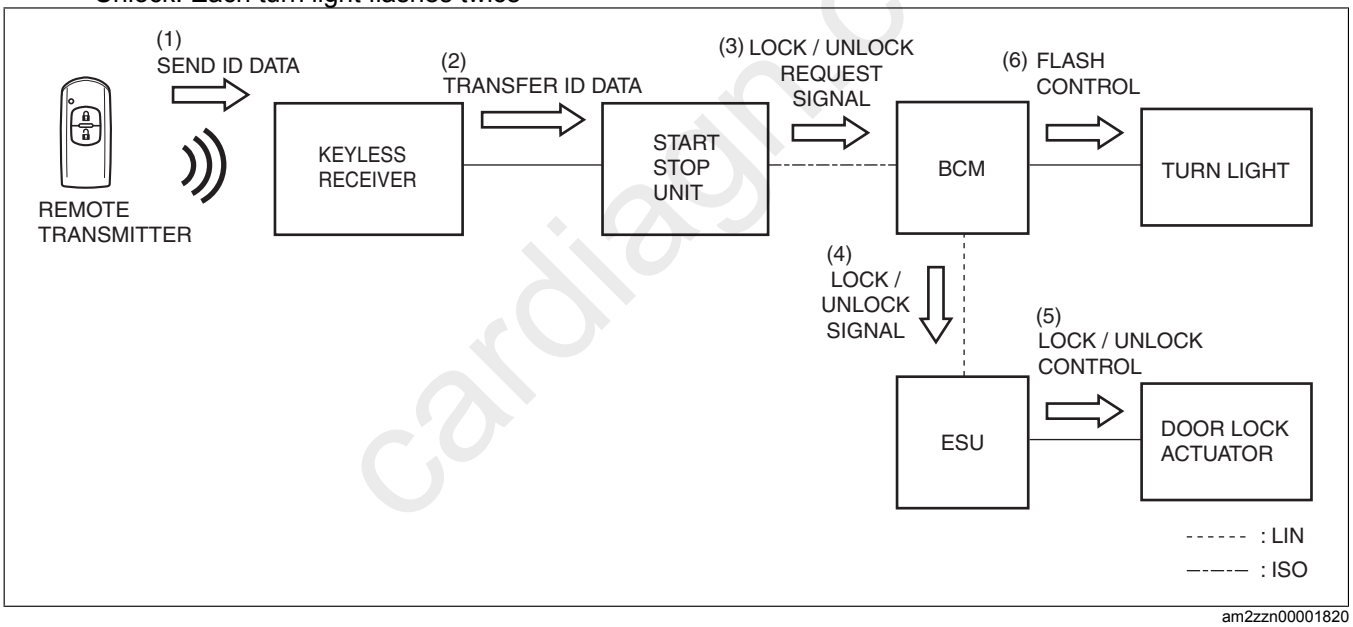
Operation
Lock/unlock operation using remote transmitter
Operation Conditions

- When door locking is performed
  - — Lock operation is performed when the lock button of the remote transmitter is operated with all of the following conditions met.
    - ACC relay and IG1 relay OFF (ignition switched OFF) detected
    - All door latch switch OFF (all doors closed condition) is detected
    - Liftgate latch switch OFF (liftgate closed condition) is detected (5HB)*1

*1 : The operation conditions are included depending on the vehicle's period of manufacture.

- When door unlocking is performed
  - — When the unlock button of the remote transmitter is operated with the ACC relay and IG1 relay OFF (ignition switched OFF) detected, unlock operation is performed.

1. When the lock or unlock button of the remote transmitter is pressed, the remote transmitter sends (1) ID data.
2. The keyless receiver receives ID data and transmits (2) the data to the start stop unit.
3. The start stop unit verifies the ID data, and if the ID data matches, it sends (3) a lock/unlock request signal to the BCM.
4. When the BCM receives the lock/unlock request signal, it sends (4) a lock/unlock signal to the electrical supply unit (ESU) via LIN communication.
5. When the electrical supply unit (ESU) receives the lock/unlock signal, it operates (5) the front/rear door lock actuator motor to lock/unlock the front/rear doors.
6. The BCM flashes (6) each turn light as follows:
   - Lock: Each turn light flashes once
   - Unlock: Each turn light flashes twice



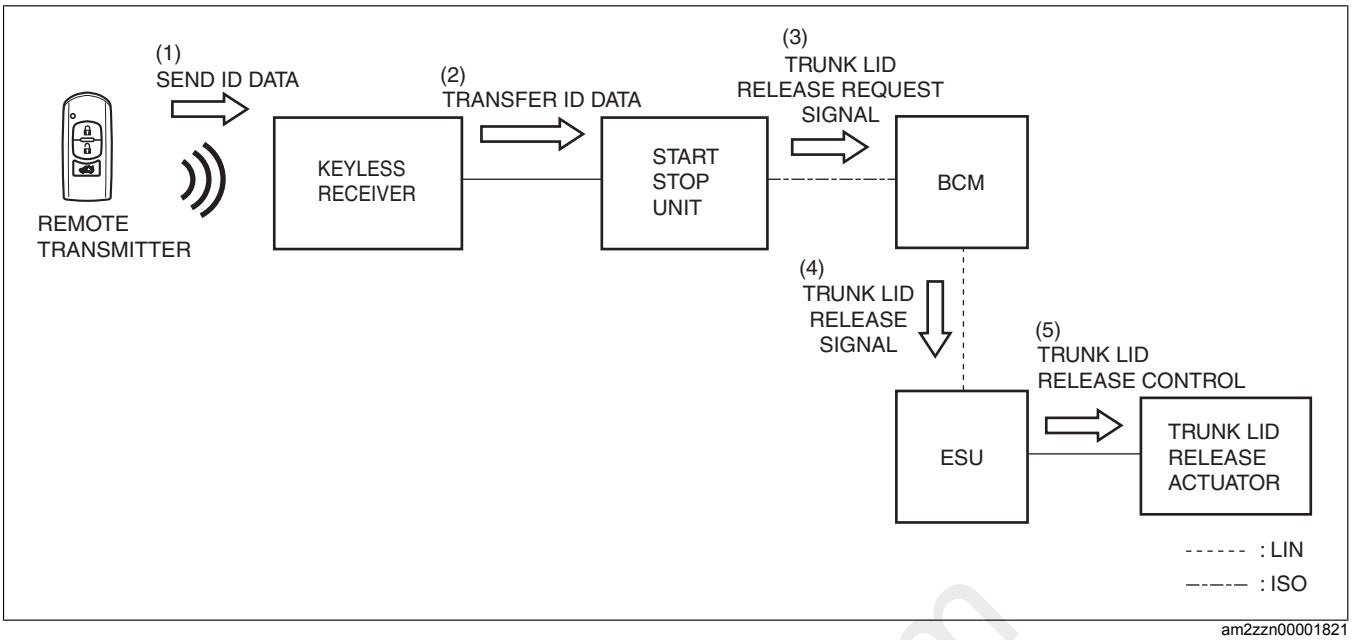
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Trunk lid opening operation using remote transmitter
Operation Conditions

- The trunk lid opens when the trunk lid open button of the remote transmitter is operated with the ACC relay and IG1 relay off (ignition switched OFF condition).

1. When the trunk lid open button is pressed for **approx. 0.5 s**, the remote transmitter sends (1) the ID data to the keyless receiver.
2. The keyless receiver transmits (2) the received ID data to the start stop unit.
3. The start stop unit verifies the ID data, and if the ID data matches, it sends (3) a trunk lid release request signal to the BCM.
4. When the BCM receives the trunk lid release request signal, it sends (4) a trunk lid release signal to the electrical supply unit (ESU) via LIN communication.
5. When the electrical supply unit (ESU) receives the trunk lid release signal, it operates (5) the trunk lid release actuator motor to open and opens the trunk lid.

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Auto re-lock function operation

1. After the start stop unit sends an unlock request signal to the BCM, if any of the following conditions is not met for 30 s[*1], it sends a lock request signal to the BCM.
   - — Any door latch switch or liftgate/trunk lid latch switch on signal (any door or liftgate/trunk lid open) received
   - — Front door lock-link switch lock (driver's door locked) detected
   - — ACC relay or IG1 relay ON (ignition switched to ACC or ON) detected
2. When the BCM receives a lock request signal, it sends a lock signal to the electrical supply unit (ESU) and locks the front/rear doors.

*1 : Personalization setting time

Remote transmitter pause function operation

When door locking is performed

- If a door is locked by the remote transmitter operation when all the following conditions are met, all the functions of the remote transmitter inside the vehicle, other than the remote transmitter used for the door lock operation are paused.
  - — All door latch switch and liftgate/trunk lid latch switch on signals (all doors, liftgate/trunk lid closed) received
  - — ACC relay and IG1 relay OFF (ignition switched OFF) detected

When the liftgate/trunk lid is closed

- When the liftgate/trunk lid latch switch off signal (liftgate/trunk lid closed) is received with all the following conditions met, all the functions of the remote transmitter inside the vehicle are paused.
  - — Front door lock-link switch lock (driver's door locked) detected
  - — All door latch switch off signals (all doors closed) received
  - — ACC relay and IG1 relay OFF (ignition switched OFF) detected

Remote transmitter pause cancel operation

- The functions of the paused remote transmitter are restored when any one of the following conditions is met:
  - — Remote transmitter with which engine can be started is brought inside vehicle
  - — Lock/unlock button on paused remote transmitter is pressed

Alert function operation

- Under the conditions in the table below, the start stop unit sends a security warning request signal via CAN transmission to the instrument cluster.
- The instrument cluster turns on/flashes the KEY warning light (red) when it receives the security warning request signal.

Condition	Instrument cluster	
	Keyless warning alarm	KEY warning light (red)
Keyless entry system malfunction[*1]	—	On
The driver's front door is opened with the ignition in ACC.	Pattern A[*2] Continuous	—
The remote transmitter cannot be detected inside the vehicle when all doors are closed with the ignition not switched OFF.	Pattern B[*2]	Flash

*1 : For vehicles with the center display, a warning message in the center display is displayed when the KEY warning light (red) is turned on. For the message content and verification method, refer to the [CENTER DISPLAY]. (See CENTER DISPLAY [WITH CENTER DISPLAY].)
*2 : For the keyless warning alarm sound pattern, refer to the [KEYLESS WARNING ALARM]. (See KEYLESS WARNING ALARM.)

Remote transmitter battery voltage low indication function operation

- When the ignition is switched from ON (engine off or on) to ACC or OFF under the condition indicated in the table, the start stop unit sends a battery voltage low indication request signal to the instrument cluster as a CAN signal.
- The instrument cluster flashes the KEY indicator light (green) when it receives the battery voltage low indication request signal.

Condition	Instrument cluster
	KEY indicator light (green)
Start stop unit receives code for low remote transmitter battery voltage*1	Flash (for **approx. 30 s** after ignition switched from ON to OFF or ACC)

*1 : If the remote transmitter battery power is 0, the remote transmitter battery voltage low indication function does not operate.

- For vehicles with the center display, a warning message in the center display is displayed when the KEY indicator light (green) flashes. For the message content and verification method, refer to the [CENTER DISPLAY]. (See CENTER DISPLAY [WITH CENTER DISPLAY].)

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